

Does Adding Painful Information to Tactile Stimuli Improve the Accuracy of Perceived Locations?

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Abstract

This study investigated whether localization accuracy could be improved by combining painful and tactile stimuli, exploring that this would provide insight into the mechanisms responsible for the accuracy of tactile information processing. Pain stimulation devices with small pin sizes delivered tactile and painful stimuli by pressing against the left palm. Ten college students participated in the experiment and reported the intensity of the pain sensation and perceived locations. The intensity of the pain sensation increased significantly with smaller pin sizes but did not vary by stimulated locations. Length of constant error (CE), the distance between the stimulated and perceived locations, did not seem to change with pin sizes, implying that the CE length did not change with the intensity of the pain sensation. These results indicated that it is difficult to simply integrate nociceptive and tactile signals in the brain, which were thought to contribute to the accuracy of perceived locations.

CCS Concepts

• **Human-centered computing** → Interaction design; Empirical studies in interaction design.

Keywords

Passive touch, tactile information transfer, pain stimuli

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1 Introduction

In recent years, there has been growing interest in tactile devices that transmit information through touch, finding use in various applications like driving assistance and virtual reality [3, 6, 9]. One useful aspect of tactile information transmission is that tactile navigation has been shown to reduce workload compared to visual displays [8]. However, previous studies on tactile sensation have reported that individuals tend to misperceive the locations of tactile stimuli [5]. Attempts to reduce this constant error (CE) length, the distance between the stimulated and the perceived locations, using vibration stimuli didn't yield significant improvements [7]. Typically, tactile stimuli transmit to the brain via $A\beta$ afferent fibers, while pain transmits via $A\delta$ or C fibers [1]. This study investigated whether adding pain stimuli to tactile stimuli, thereby activating both $A\beta$ and $A\delta$ fibers, could improve the accuracy of perceived locations. Akamatsu [1] demonstrated that smaller diameter cylindrical stimulators induced lower pain thresholds. Therefore, pain stimulation devices with different pin sizes were designed in this study. Considering safety, the minimum pin size for the created device was set to $\Phi 1$, and devices with $\Phi 3$ and $\Phi 5$ pins were also developed and experimentally evaluated to assess their impact on the accuracy of perceived locations.

2 METHODS

The experiment was conducted on ten right-handed male university students aged 21–24. All participants were informed about the experiment, including their right to withdraw due to excessive pain, and provided written consent before participating.

The XY plotter was modified by attaching a push solenoid to the head section for the experiment. The equipment was constructed to send stimuli to any locations via an Arduino board connected to a PC's serial port (Figure 1, left). A web camera was installed at the bottom of the equipment to record the participant's palm. Pain stimulation devices with different pin sizes (Figure 1, right) were attached to the tip of the push solenoid. Additionally, an acrylic plate with a 90 mm $\times$ 76 mm opening was placed on top of the equipment.

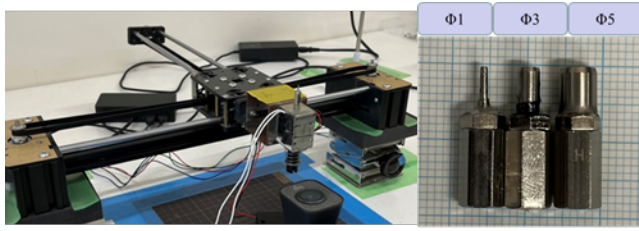


Figure 1: Equipment for the Experiment (Left) and Pain Stimulation Devices (Right)

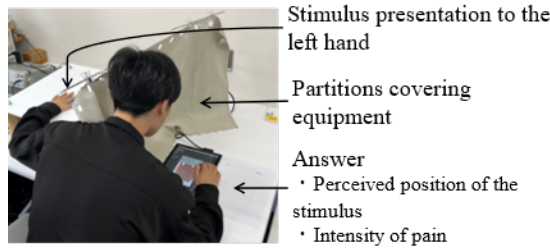


Figure 2: Experimental Environment

The objective of this study was to evaluate whether adding painful information to tactile stimuli improve the accuracy of perceived locations.

Figure 2 shows the experimental environment. Participants were seated, placed their left palm on the equipment, and responded to perceived locations and the intensity of the pain sensation using their right hand. Partitions were placed so that participants could not see the stimulated locations.

The experiment was conducted after participants had experienced the intensity of the pain sensation using the pin size of $\Phi 1$ and confirmed that they were able to continue the trials. The opening was divided horizontally and vertically into equal sections, and six locations were set and numbered from their intersections, and stimulus presentation was made at these locations (See Figure 3). Routes were decided that would stimulate these six locations. Considering order effects, a total of nine conditions were set for the experiment by combining three pin sizes with three stimulation routes. Participants indicated perceived locations on an image of their palm projected on the tablet and rated the intensity of the pain sensation using a Visual Analog Scale (VAS), which ranged from "no pain" to "maximum imaginable pain."

A two-way ANOVA was performed with pin sizes and stimulated locations as independent variables, and the intensity of the pain sensation and CE length as dependent variables. This study also confirmed a correlation between CE length and the intensity of the pain sensation. Statistical analysis was performed using R 4.4.1.

3 RESULTS

The correlation coefficient between the intensity of the pain sensation and CE length for all data was 0.0429 ± 0.0846 ($t=0.996$, $p=0.316$).

The intensity of the pain sensation varied significantly based on each pin size ($F(2,522)=108$, $p<0.001$). The pin size of $\Phi 1$ resulted in

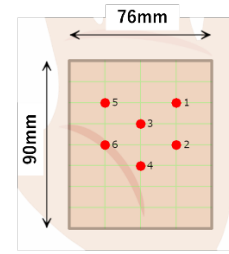


Figure 3: Locations for Stimulus Presentation

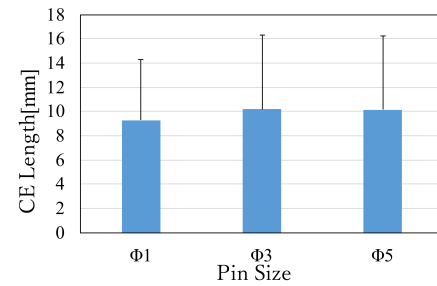


Figure 4: Relationship between Pin Size and CE Length

the largest the intensity of the pain sensation (30.5 mm), while the pin size of $\Phi 5$ yielded the smallest the intensity of the pain sensation (9.14 mm). There was no significant difference in the intensity of the pain sensation depending on the stimulated locations ($F(5,522)=1.22$, $P=0.295$).

The CE length varied significantly between stimulated locations ($F(5,522)=5.51$, $p<0.001$). Location 1, which close to the base of the little finger, showed the smallest CE length (mean 8.18 mm), while location 4, which close to the wrist, showed the largest CE length (mean 11.5 mm). However, the CE length did not change based on each pin size ($F(2,522)=1.48$, $P=0.228$) (See Figure 4).

4 RESULTS

The results of this experiment showed that the CE length for each stimulated location was similar to that reported in previous studies [7]. However, although the reduction in pin size had a significant effect on the intensity of the pain sensation, there was no improvement in the accuracy of perceived locations. According to a previous study [4], pain stimuli were transmitted to the somatosensory cortex, where the locations of the pain were identified. In light of this finding, it was possible that the decrease in pin size, while producing pain, also introduced an additional obstacle to accurate stimulus localization. One hypothesis was that the smaller deformable area of the palm skin reduced the number of activated receptors, resulting in decreased stimulus intensity and, consequently, lower accuracy of perceived locations. Indeed, previous studies had shown that stimuli of low intensity could cause ambiguity in perception [2]. In other words, although pain stimuli were applied, it was possible that the total amount of stimulus information had not been adequately controlled in this experiment. Therefore, it may have been necessary to conduct further experiments in which

factors such as stimulus intensity and the area of skin deformation were systematically controlled. Further elucidation of the human tactile perception system, from receptors to the cerebral cortex, is also needed to improve the accuracy of perceived locations. The finding that pain information was not valuable for the accuracy of perceived locations in this study provides an important insight for future guidelines on the development of passive devices.

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